

Automotive Edge-Cloud Orchestration Report

Project Overview

Self-Adaptive Task Orchestration using Machine Learning and Deep Reinforcement Learning for Vehicle-Edge-Cloud environments.

System Metrics

Metric	Value
Average Reward	17.45
Average Latency	152.48 ms
Average Battery	55.73 %

Offloading Distribution

Target	Percentage
Vehicle Processing	33.0 %
Edge Offloading	37.0 %
Cloud Offloading	30.0 %

Executive Summary

The Hybrid ML + DQN Orchestrator achieved the best overall performance by balancing latency, energy efficiency, and adaptive workload offloading across Vehicle, Edge, and Cloud resources.

Best Strategy Selected
Hybrid Orchestrator
Reason
Lowest average latency
Highest reward score
Highest decision accuracy

Best energy efficiency

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